Agenda

01 **Framework**
Inputs, outputs, and the resource-allocation process

02 **Classic response-model example**
CallPlan allocation and field validation

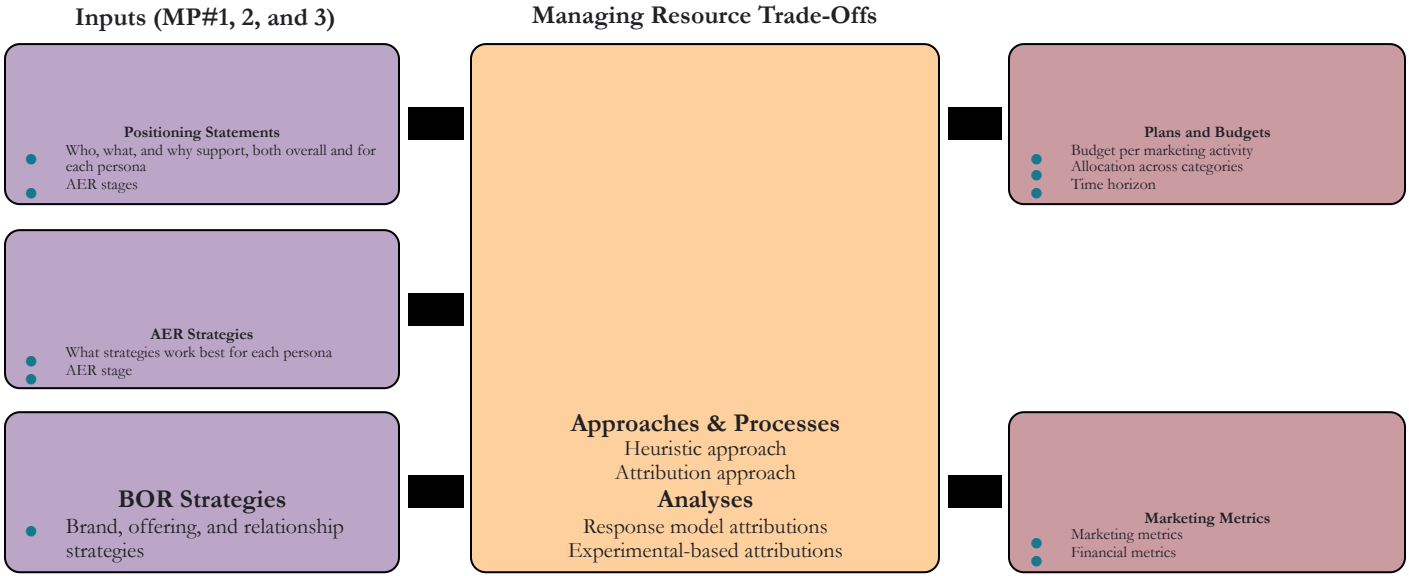
03 **AI Frontier case workshop**
Double Machine Learning for causal marketing allocation

04 **Takeaways**
Attribution, optimization, metrics, and validation

Framework for Managing Resource Trade-Offs

- Inputs to the Managing Resource Trade-Offs Framework
  - → Positioning Statements
  - → AER Strategies
  - → BOR Strategies
- Outputs of the Managing Resource Trade-Offs Framework
  - → Plans and Budgets
  - → Marketing Metrics

Marketing Principle #4: All Resources Are Limited → Managing Resource Trade-Offs



Inputs to the Managing Resource Trade-Offs Framework

- The outputs of the three previous Marketing Principles serve as the key inputs to the resource trade-off framework, such that each MP requires some initial trade-off decisions
- The positioning statement attained through MP#1 answers key questions about who customers are, what needs the product or service can fulfill, and why this product or service is the best option to satisfy those customers' needs
- The AER positioning statements, the output from MP#2, also are key inputs for the resource trade-off framework
- Finally, the input derived from MP#3, which builds on MP#1 and MP#2, describes how to use BOR strategies to build SCA and erect strong barriers against competitive attacks

Outputs of the Managing Resource Trade-Offs Framework

- A fundamental problem for effective resource allocation is identifying and measuring the best or most appropriate metrics
- As a popular saying holds, a firm is only likely to achieve what it measures
- For most marketing resource investments, both financial and marketing metrics are necessary to capture the different aspects of the benefits earned from the investment

Outputs of the Managing Resource Trade-Offs Framework

- Another set of outputs pertains to the three components of each resource allocation decision:
  - → **Budget per marketing activity**, or the size of the commitment the firm makes to the marketing activity
  - → **Allocation across categories**, which reflects the percentage split of the marketing budget for a specific activity across categories
  - → **Time horizon** of the budget, involving the timespan for which the firm commits to this marketing budget
- Thus when choosing its advertising budget for example, a firm would determine how many total dollars to spend (budget) on different forms of advertising (e.g., print and online), as well as how long to run the advertising campaigns (e.g., two months)

Example: KIND (US)

- The relatively young and small US healthy snack company struggled to allocate its limited resources.
- Initially, CEO David Lubetzky invested too many resources in expansion efforts and other immediate growth-oriented endeavors, only to learn they were not in the company's best interest.
- Therefore, he began diverting more resources to R&D. Famously, KIND spent two years developing its dark chocolate and sea salt bar.
- As a result of this reallocation of resources, investors poured millions of dollars into the firm. Today, KIND products are on tens of thousands of grocery store shelves across the United States and internationally

Process for Managing Resource Trade-Offs

- Five-step process for managing resource trade-offs:
  - → Identify strategically relevant metrics
  - → Assess the relationship between metrics and marketing resources
  - → Assess the optimality of the resource allocation decisions
  - → Finalize the resource allocation decisions
  - → Integrate across different marketing activities

Example: Chevron (US)

- This US energy corporation goes to great lengths to ensure it makes the right decisions, especially costly ones.
- Chevron established Decision Quality (DQ) standards, which require analyses of alternative outcomes, occurrence probabilities, and their results to define the best decision.
- In 2000, former CEO David O'Reilly made DQ mandatory for all capital expenditure decisions worth more than $50 million.
- Doing so has proven successful for Chevron. In one instance, it managed to capture 50% of the value at 25% of the cost for a potentially $2 billion expenditure.

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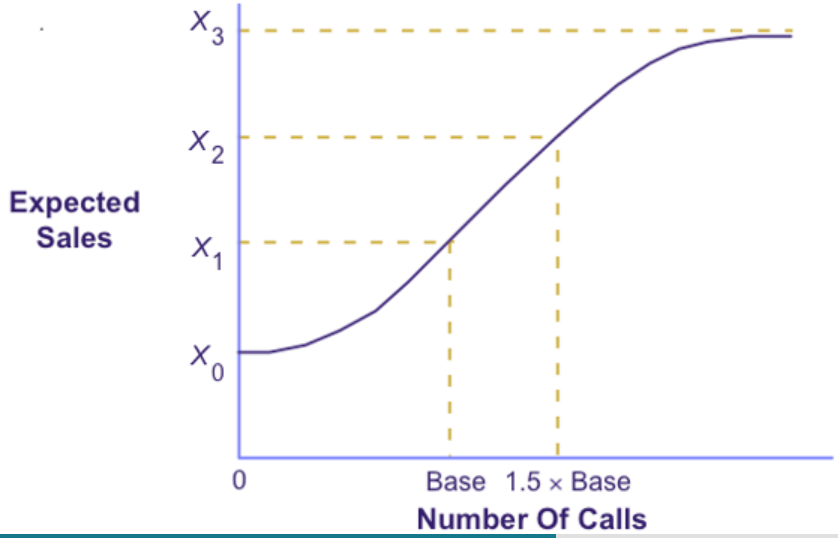
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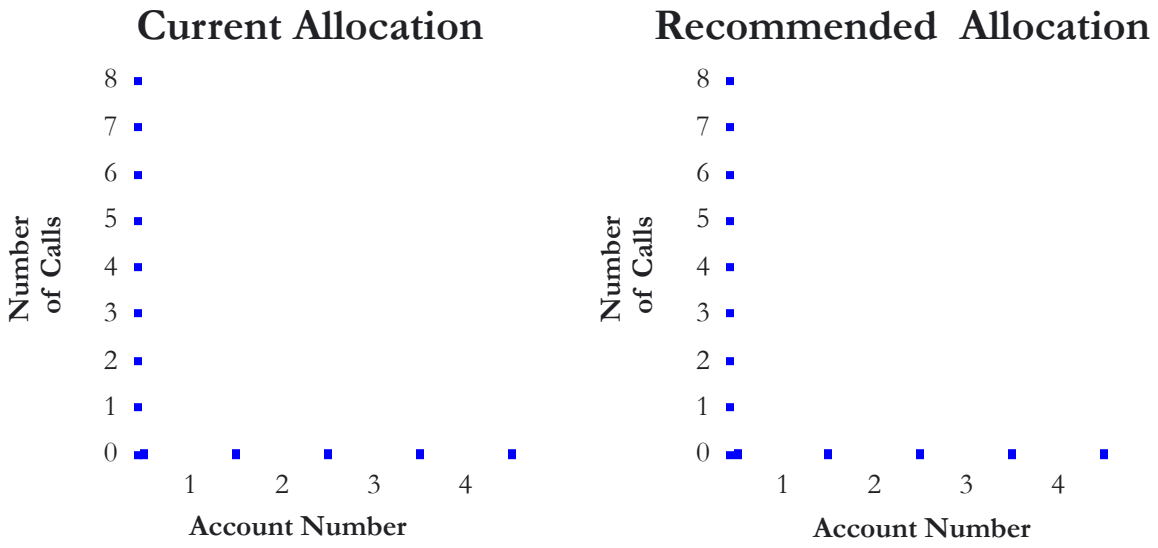
Real Example: CallPlan Response Model at United Airlines

- **Objective:** Allocate salesperson time to maximize gross profits
- **Build and calibrate model** from average salesperson answer to 4 questions: What would sales of this account be if . . .
  - → You don't visit accounts (X0)
  - → You visit the account the same as last period (X1)
  - → You visit the account 50% more than last period (X2)



- **Optimize response model:**
  - → Maximize sum of account revenues x account margin
  - → Constraints are salesperson time and min./max. calling restriction
  - → Provides recommended calling frequencies for each account

Current Allocation vs. Response Model's Recommended Allocation



- Profit with current allocation: $11,985
- Predicted profit with optimal allocation: $13,000
- Do you believe this will happen?

Test Recommended Allocations in Field Experiment

- Sample
  - → 20 randomly selected geographically-stratified salespeople
  - → 20 other salespeople matched on key demo (age, experience, territory)
  - → Randomly selected 1 from each pair, others were control
- Treatment
  - → Both groups were asked judgment questions, but only one group given recommended allocations
  - → Other group given computer feedback summarizing answer to questions (helps prevent a demand effect)
- Evaluated salesperson sales growth after six months
  - → Experimental group 11.9%
  - → Control group 3.8%
  - → Difference 8.1%, probability this is due to chance is < .025

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AI Frontier: Double Machine Learning for Resource Allocation

Chapter 8 asks how much each marketing resource contributes to performance. Double Machine Learning combines flexible prediction with causal estimation so that complex controls do not dominate the effect of interest.

- **Target:** the incremental effect of a marketing input on contribution profit.
- **Nuisance functions:** how observed conditions predict both spending and outcomes.
- **Decision:** use causal marginal returns as inputs to a constrained allocation model.

Classic response model

Models outcome as a function of spending and controls.

→

DML extension

Uses cross-fitting and an orthogonal score to reduce bias from flexible nuisance models.

Sources: Chernozhukov et al. (2018), The Econometrics Journal; DoubleML documentation.

Case Workshop: SkyLink Airlines

Managerial question

SkyLink wants to reallocate paid-search spending across regional markets without increasing its total budget.

Why the historical pattern is difficult to interpret

Managers spend more in markets where demand, capacity, and promotions are already high. These same factors also increase contribution profit.

36
regional markets

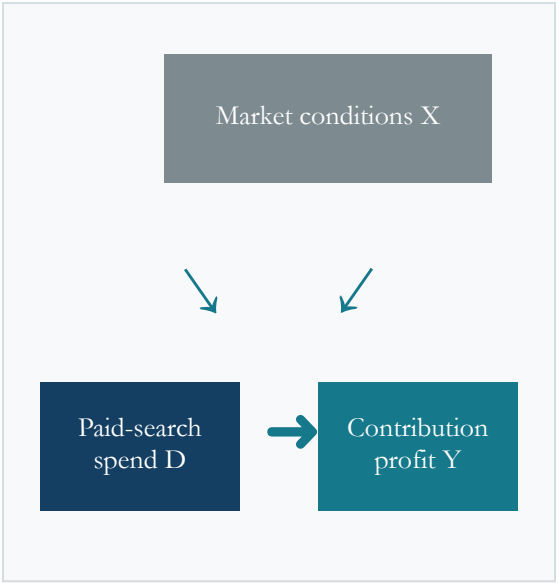
52
weeks

1,872
market-week observations

3
market tiers

The supplied teaching dataset is synthetic. It reproduces a realistic confounding problem without representing any actual airline.

Why Prediction Is Not Attribution



Observed association

Markets with high demand receive more advertising and generate more profit. A predictive model may attribute part of the demand effect to advertising.

Causal target

Estimate how profit would change if paid-search spend changed while comparable market conditions remained fixed.

In the workshop data, the unadjusted spend-profit slope is 9.07. That number is far too large to use as an incremental return.

The Partially Linear Model

$$Y = \theta D + g(X) + \varepsilon$$

$$D = m(X) + v$$

Y	Contribution profit
D	Paid-search spending
X	Demand, fares, capacity, promotions, competition, seasonality, and prior bookings
θ	Causal contribution profit per additional dollar of paid-search spending

Machine-learning models estimate g(X) and m(X). DML protects the estimate of θ from first-order nuisance-model errors through orthogonalization and cross-fitting.

Model and notation follow Chernozhukov et al. (2018) and the DoubleML PLR documentation.

Cross-Fitting and Orthogonalization

01

Split the sample

Divide market-week observations into five folds.

02

Learn nuisance functions

On the training folds, predict profit from X and spending from X.

03

Residualize held-out data

Remove the predicted components from Y and D in the held-out fold.

04

Estimate the focal effect

Regress residualized profit on residualized spend and combine results across folds.

$$[Y - \ell(X) - \theta(D - m(X))] \times [D - m(X)] = 0$$

DoubleML uses Neyman-orthogonal scores and cross-fitting to reduce regularization and overfitting bias.

Workshop Data and Analysis

Files

Dataset: 1,872 market-week observations

Python: transparent five-fold DML implementation using random forests

Analysis sequence

1. Describe spending and profit by market tier.
2. Estimate the naive spend-profit slope.
3. Estimate the overall DML effect.
4. Repeat DML within Core, Growth, and Emerging markets.

Outcome Y

Contribution profit

Treatment D

Paid-search spend

Controls X

Market demand, price, capacity, promotions, competition, fuel cost, seasonality, and prior bookings

Workshop Results

Unadjusted slope

9.07

→

DML estimate

1.36

95% CI: 1.16 to 1.57

Market tier	DML effect	95% confidence interval	Allocation implication
Growth	1.84	1.49 to 2.20	Highest priority for incremental budget
Core	1.51	1.16 to 1.87	Positive return, subject to saturation
Emerging	1.18	0.82 to 1.54	Lower priority and greater uncertainty

Illustrative results produced by the supplied synthetic dataset and Python script. Effects represent contribution-profit dollars per additional dollar of paid-search spending.

From Causal Response to Budget Allocation

1

Estimate

DML estimates causal marginal returns overall and by market tier.

2

Optimize

Allocate the fixed budget toward higher-return markets while respecting capacity, minimum-presence, and risk constraints.

3

Validate

Randomly implement the recommended reallocation in selected markets and measure incremental contribution profit.

Connection to the Chapter 8 framework

Inputs: positioning, lifecycle, and competitive priorities determine which controls and constraints matter.

Outputs: activity budgets, allocation across markets, time horizon, and the metrics used for evaluation.

Case Workshop Deliverable

Team recommendation

- State which market tier should receive the first incremental budget and why.
- Specify at least two business constraints for the allocation model.
- Design a field experiment that tests the recommended shift.

Required caveats

- DML requires observed controls that make treatment assignment conditionally independent of potential outcomes.
- Poor overlap limits what the data can identify.
- A historical causal estimate may change when the market or allocation policy changes.

What evidence would make you comfortable moving from the DML recommendation to a permanent budget change?

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Takeaways

- All resources are limited. Managers must manage resource trade-offs to develop an effective marketing strategy. Most marketing decisions require trade-offs across multiple objectives, because resources are constrained and often interdependent.

- Several factors increase the need for ongoing resource trade-offs, including limited resources (resource slack), changes in the composition of consumer segments, changes in the lifecycle stages of the product portfolio, changes in the market landscape due to competitive actions, and changes in the effectiveness of marketing activities.

- Approaches to managing resource trade-offs have evolved from an exclusively heuristic-based era, in which managers solved resource allocation problems using simple rules of thumb, intuition, and judgment, to a data-based era, in which managers rely on statistical models and detailed information.

- The heuristics approach relies on anchors, often related to spending in the previous period, which managers use to make marketing resource allocation decisions. Then managers may adjust their decisions every period, after observing the prior outcomes.

Takeaways

- An attribution approach asks, How does a specific (e.g., 1 percent) increase in a resource option affect a particular outcome, keeping all else constant? The model integrates past decisions and past outcomes, then produces a mathematical assessment of how much impact each resource trade-off truly has for generating outcomes.

- A response model-based attribution approach captures the relationship between past marketing resources and past outcomes. A basic assumption is that past outcomes relate to future outcomes, which is usually reasonable. The use of past data then can uncover the relationship between marketing resources and performance.

- There are three key inputs and two key outputs of the framework for managing resource trade-offs. The inputs are the outputs of the previous three principles. The outputs are a description of the firm's resource plans and budgets and the use of key marketing metrics that can effectively validate these resource outlays.

Readings

- *Note on Marketing Performance Assessment* (great overview on marketing metrics and process for using metrics)
- *The Rules of Measurement* (process for NPS, response rate and bias)
- *Building Loyalty in Business Markets* (building B2B loyalty)
- *Marketing Strategy Book*: Chapter 8